Solid Mass Diffusivity

proportionality constant →

$$D_{AB} = D_o e^{-Q/RT}$$

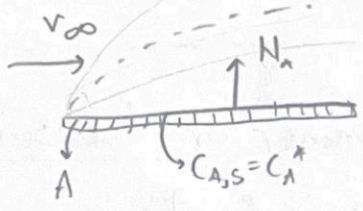
Convective Mass Transfer

Mass transfer of a solute into a moving fluid:

$$N_A = k_c (C_{A,S} - C_{A,\infty})$$

k_c → convective mass transfer coeff

When any moving fluid encounters a solid ⇓ Boundary layer (Both velocity and conc. BL)



$\therefore -N_A \cdot S = \frac{dm_A}{dt}$ (Mass of A decreases until all is 0)

$\Rightarrow N_A = \frac{m_{A_0}}{t \cdot S}$

$\Rightarrow \boxed{k_c (C_{A,S} - C_{A,\infty}) = \frac{m_{A_0}}{t \cdot S}}$

$$S_c = \frac{\text{momentum diffusivity}}{\text{Mass diffusivity}} = \frac{\mu/\rho}{D_{AB}} = \frac{\nu}{D_{AB}}$$

(ν: kinematic viscosity; momentum diffusivity: diffusion due to pressure, shear stress)

↑ Schmidt number

↳ physically relates the thickness of hydrodynamic layer & mass transfer boundary layer

$$Sh = \frac{\text{Molecular ME resistance}}{\text{Convective MT resistance}} = \frac{k_c L}{D_{AB}}$$

+ Laminar Flow in BL

Using mass balance and BC's, we arrive at: (and Blansius approx.)

$$\left.\frac{dC_A}{dy}\right|_{y=0} = (C_{A,\infty} - C_{A,S}) \left(\frac{0.332}{x} Re_x^{1/2}\right)$$

as $N_{A,y} = -D_{AB} \left.\frac{\partial C_A}{\partial y}\right|_{y=0} = k_c (C_{A,S} - C_{A,\infty})$

(for $S_c = 1$)

we get, k_c is a function of x

$$k_{c,x} = \frac{D_{AB}}{x} [0.332 Re_x^{1/2}]$$

or $$\boxed{\frac{k_{c,x} \cdot x}{D_{AB}} = Sh_x = 0.332 Re_x^{1/2}}$$

When $Sc \neq 1$: $\frac{\delta}{\delta_c} = Sc^{1/3}$ (hydrodynamic BL / concentration BL)

&

$Re_x < 2.0 \times 10^5$ [Laminar]

$$\left.\frac{\partial c_A}{\partial y}\right|_{y=0} = (c_{A,\infty} - c_{A,s})\left(\frac{0.332}{x} Re^{1/2} Sc^{1/3}\right)$$

$$\therefore \boxed{\frac{k_{c,x}\, x}{D_{AB}} = Sh_x = 0.332\, Re_x^{1/2} Sc^{1/3}}$$

We can also determine the average mass-transfer coefficient:

$$k_c = \frac{\int_0^L k_{c,x}(x)\, dx}{\int_0^L dx} = 0.664 \frac{D_{AB}}{L}\left(\frac{v_\infty L}{\nu}\right)^{1/2} Sc^{1/3} = 2\, Sh_x\big|_{x=L}$$

($\frac{v_\infty L}{\nu}$ = Re)

However, when the flow is not laminar or the surface is not a flat plate, then we use Von Kármán approximation.

